

Spiking keyword detection on PicoRV32

Group 2 | Keyword: yes

Speech Commands v0.02 | PYNQ-Z2

GPU training, integer deployment, RTL verification

Board: 40/40 vectors bit-exact (loaded over JTAG)

Question and hypothesis

Can a small SNN fit 256 KiB BRAM and finish within a 250 ms hop?

Hypothesis: caching the constant projection cuts latency without a large F1 penalty.

Compare current and deterministic rate encoding under equal training budgets.

Controlled experiment

Official speaker-disjoint splits: 84,843 / 9,981 / 11,005

Yes vs all other 34 words

768 inputs -> 64 LIF neurons -> 2 linear readouts

12 steps, beta 7/8, immediate subtractive reset

3 seeds x 2 encodings x 35 epochs; final 10 use QAT

Checkpoint and threshold selection use validation data only.

Training and deployment share a frontend

16 kHz, one-second mono audio

400-sample Hann / 160 hop / 512 FFT

24 mel bands x 32 time bins -> 768 uint8 bytes

Signed Q10 int16 weights and int32 accumulators

Independent NumPy, native C, and RV32IM implementations

No floating point on the RISC-V processor.

Detection results

Deployment: current_seed2

Precision 90.15% | Recall 85.20% | F1 87.61%

TP 357, FP 39, FN 62, TN 10547

Background probes: 0 / 398 detections

Clip metrics do not establish continuous-speech false accepts/hour.

Latency determines the demo model

Current: max 59.82 ms

Rate: max 715.52 ms

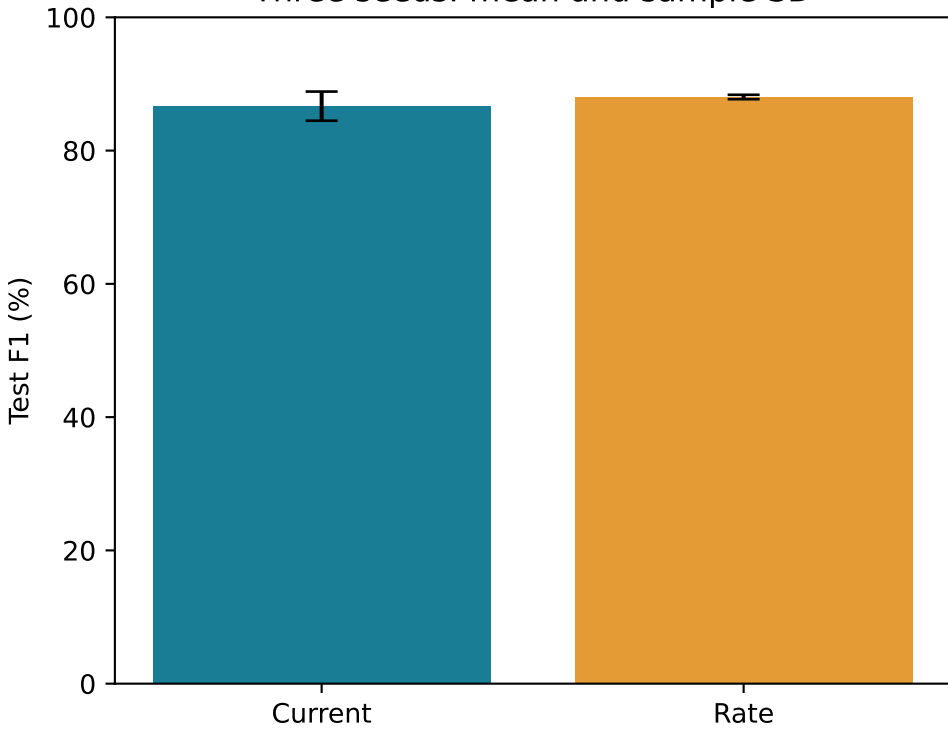
Mean speedup: 9.95x

Rate wins validation F1 but misses the 250 ms hop.

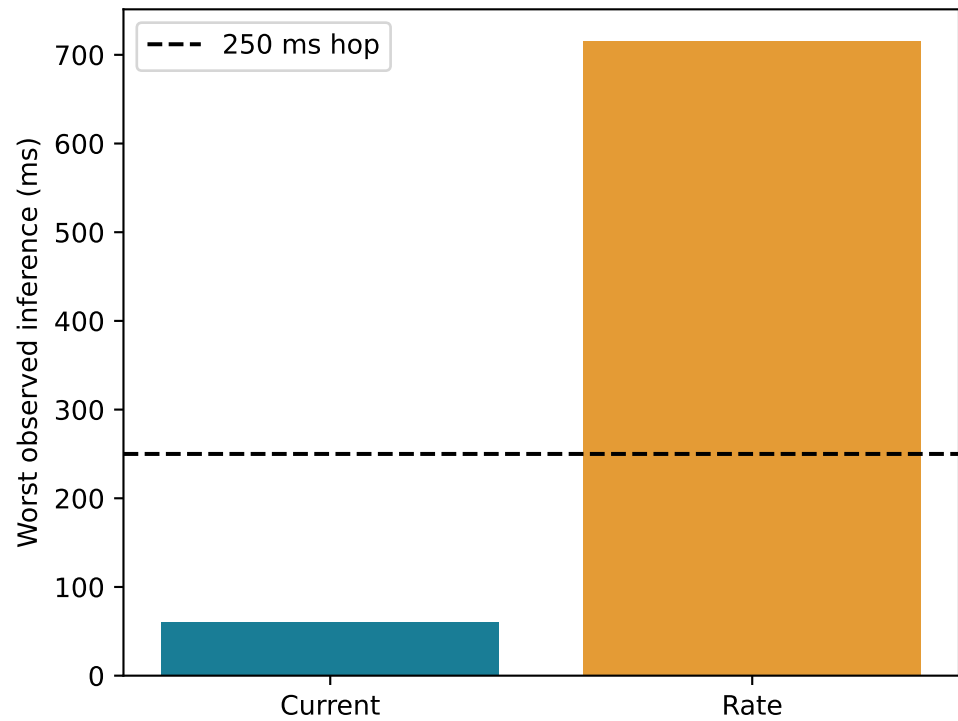
Select the feasible current model by validation F1.

Timing excludes audio-window accumulation, PC frontend and Ethernet.

Three seeds: mean and sample SD



40 RTL vectors at 100 MHz



Verification evidence

11,005 clips: native production C matches integer oracle exactly

40 vectors per encoding: actual PicoRV32 execution matches exactly

Traps, timeouts, malformed requests, and duplicate sequence checks

AXI backpressure, byte lanes, independent AW/W, read/write contention

100,000,000-cycle LED pulse; retrigger/reset/cancel/wrap checks

Real-model local TCP replay and shared WAV frontend tests

FPGA implementation

Vivado 2025.2 | XC7Z020 | 100 MHz

Setup slack +0.517 ns; hold slack +0.031 ns

2699 LUTs; 2588 registers

64 BRAM36 tiles; 6 DSPs

98,824-byte integer model; 16 KiB reserved stack

Bitstream generated; no implementation errors or critical warnings.

Demo and limitations

PC microphone -> mel features -> TCP -> ARM mailbox -> PicoRV32

Positive inference retriggers hardware LED0 for one second.

Board inference verified over JTAG (SD boot fails on this board).

Not yet measured: Ethernet path, calibrated microphone accuracy, power.

Later: kdot instruction, 2-of-3 confirmation (JOURNAL.md).

About 15% of isolated yes clips are missed.

No general energy or SNN-vs-ANN claim.

Artifacts, hashes, raw reports, and reproduction commands are included.